

EXPERIMENT # 01

OBJECT:

To study the spectral characteristics of photocell and determine the Planck's Constant.

APPARATUS:

A photocell, A DC micrometer, A reversing key, a rheostat, DC power supply, filters of different colors and Halogen tungsten lamp.

THEORY:

(When light falls on certain materials like selenium etc. (which has low work function). Electrons are emitted from their surfaces. These electrons are called "*Photo-Electrons*" and the phenomenon governing such an emission is called the photo-electric effect the material is said to be "*Photo-sensitive material*".)

In a photo-cell the photo-electric effect is utilized. It is used to convert light energy into electrical energy. The emission of photoelectrons depend upon the frequency which is proportional to the energy of the incident radiation, so by changing the frequency, the energy of the incident light can be changed which can be done by taking filters of different colors(wavelength) .

According to their construction the photoelectric cells are of three types:

(i) Photo emissive cell (ii) Photo Conductive cell (iii) Photo Voltaic cell

A phototube or photoelectric cell is a type of gas-filled or vacuum tube that is sensitive to light. Such a tube is more correctly called a 'photo emissive cell' to distinguish it from photovoltaic or photoconductive cells. Phototubes operate according to the photoelectric effect: Incoming photons strike a photocathode, generating electrons, which are attracted to an anode. Thus, current flow is dependent on the frequency and intensity of incoming photons. The light wavelength range over which the device is sensitive depends on the material used for the photo emissive cathode. A cesium-antimony cathode gives a device that is very sensitive in the violet to ultra-violet region with sensitivity falling off to blindness to red light. Cesium on oxidized silver gives a cathode that is most sensitive to infra-red to red light, falling off towards blue, where the sensitivity is low but not zero

WORKING FORMULA:

According to Einstein Photoelectric Effect Equation We have

$$hf = \phi + eV_o$$

$$V_o = \frac{h}{e}f - \frac{\phi}{e}$$

OBSERVATIONS:

Position of the lamp: 25 cm

Position of the cell: 0 cm

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30/1/2025

S.No	Colour of filter	Wavelength(λ) 10^{-9} nm	Frequency(ν) Hz	Stopping Potential (V_0) Volt
1.	Red	635×10^{-9}	4.724×10^{14}	-0.43
2.	Orange	570×10^{-9}	5.263×10^{14}	-0.66
3.	Yellow	540×10^{-9}	5.55×10^{14}	-0.85
4.	Green	500×10^{-9}	6×10^{14}	-0.98
5.	Blue	460×10^{-9}	6.521×10^{14}	-1.23

CALCULATIONS:

$$\frac{h}{e} = \frac{0.98 - 0.66}{6 \times 10^{14} - 5.26 \times 10^{14}}$$

$$h = 6.918 \times 10^{-34} \text{ Js}$$

$$\phi = hf_0$$

$$\phi = 6.918 \times 10^{-34} (3.8 \times 10^{14})$$

$$\phi = 2.629 \times 10^{-19} \text{ J}$$

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$$\% \text{ age error} = \frac{\text{Observed value} - \text{True value}}{\text{True value}}$$

$$= \frac{6.918 \times 10^{-34} - 6.634 \times 10^{-34}}{6.634 \times 10^{-34}}$$

$$= 0.042 \times 100$$

$$\% \text{ age error} = 4.28\%$$

RESULT:

The value of Planck's constant is $6.918 \times 10^{-34} \text{ Js}$.

The percentage error in observed value is 4.28% .

The work function of cathode material is $2.629 \times 10^{-19} \text{ J}$.

PRECAUTIONS:

1. The light from the lamp should fall normally on the cell.
2. To avoid the over overheating the photo cell, the lamp should be switched off during the observations.
3. The connections should be tight.

Questions

Q1: Explain the construction and working of photocell.

Q2: What sensitive material is used when the cell is to be used for a visible light?

Q3: What is threshold frequency?

Q4: What is photo-electric work function?

Q5: How will you determine the stopping potential?

Q6: Does the violet light have more energy than red light?

Q7: What is the effect of intensity on photoelectric current?

Q8: What is the order of current in photocell?

Ans 6) Yes violet light have more energy than red light
bcuz energy depend upon its frequency and violet
has higher freq and shorter wavelength.

Ans 7) Increasing light intensity increases the photoelectric
current bcuz more photons eject more electrons. However,
it does not affect the K.E of electron.

Ans 8) The order of current in photocell is typically in the range
of micro-ampere to milliamperes depending upon intensity
of light and material used.

Ans 1) A photocell is a device that converts light energy into electricity using the photo-electric effect. It has a photosensitive cathode and a metal anode inside a glass tube. When light hits the cathode, it releases electrons, which flow to anode, generating electric current. The current depends on light intensity.

Ans 2) For visible light, alkali metals like Cesium and potassium are used as photosensitive material in photocell. They have a low work function, making it easier to emit e^- when exposed to visible light.

Ans 3) Threshold frequency is the min. frequency of light req. to eject e^- from the surface of a material in p.e.-effect. It depends upon the material's properties and is related to work function of material.

Ans 4) It is the min. energy needed to eject an electron from a material's surface. It is related to ν_0 .

$$\phi = h\nu_0$$

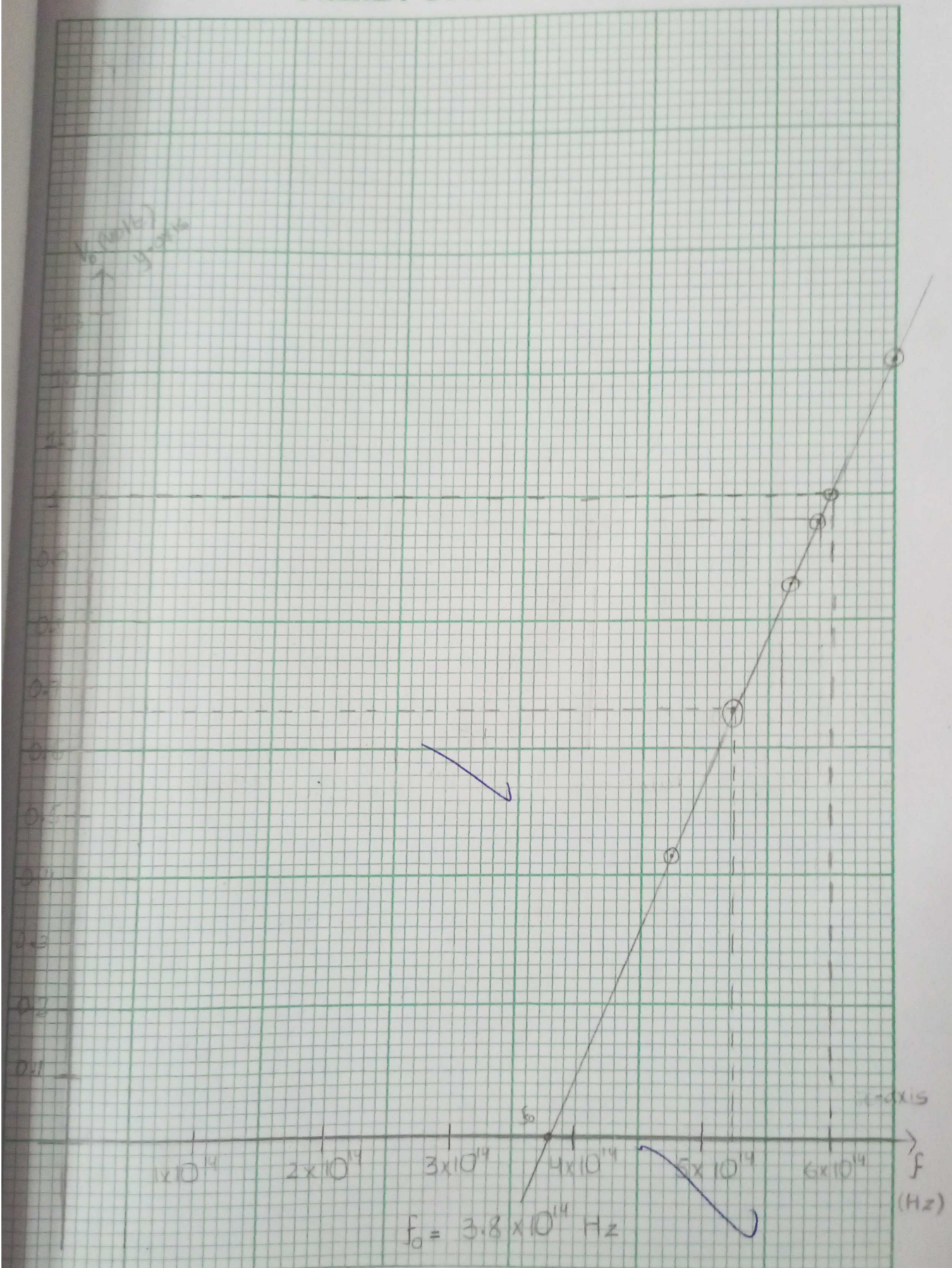
unit: eV or J

Ans 5) Stopping potential is the min negative potential applied to the anode to stop the ejected electron in p.e.-effect. It helps to find max KE of photoelectrons.

$$eV_0 = KE_{(max)}$$

$$V_0 = \frac{1}{e} (h\nu - \phi)$$

- calculate the energy of incident light
- subtract w. func.
- divide by electron charge to find V_0



EXPERIMENT # 02

OBJECT:

To find the ionization potential of mercury using a gas filled diode.

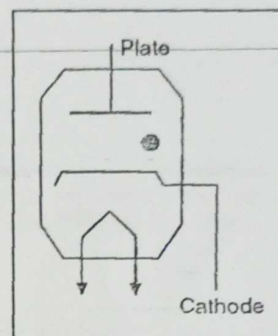
APPARATUS:

A gas filled (mercury vapors) diode, a D.C, power supply, a voltmeter, micro-ammeter, a potentiometer.

THEORY:

Ionization potential: *The least energy required to remove the most loosely bound electron from an isolated free neutral atom is known as the ionization potential of the atom.*

Ionization potential of mercury: The ionization potential of mercury can be determined by introducing mercury vapor at a low pressure of 10 mm to 50 mm of mercury column in an evacuated tube fitted with a cathode and an anode. A mercury vapor filled gas diode is the most suitable for the purpose. The cathode of the gas diode may be directly or indirectly heated type. A hot cathode gas filled diode is known as a *phanotron* [Incidentally a gas filled triode is known as a *Thyratron*] A gas filled diode is symbolically represented as shown in Fig. The dot in the tube shows the presence of the gas (or vapor).



When the anode or plate of the gas filled diode is at a positive potential with respect to the cathode, electrons move across the tube from the cathode to the anode. This electronic current depends upon two factors:

- (i) The number of electrons emitted per unit area from the cathode and its temperature.
- (ii) *The effect of space-charge* – the negatively charged region containing the electron cloud due to the accumulation of electrons emitted by the cathode.

As the plate potential is increased the plate current slowly increases. But when the plate potential is increased beyond a particular value the plate current increases. But when the plate potential is increased beyond a particular value the plate current increases much more rapidly than it does below that critical value. This is because when the plate potential approaches this critical value the electrons arriving at the anode gain enough energy to knock out the electrons from the atoms of the gas close to the anode. These electrons are also attracted by the anode causing an increase in plate current and the positive ions neutralize some of the space charge, which further helps to increase the kinetic energy of the thermo-electrons. This potential is equal to the ionization potential of the gas (or vapor) and for this value of plate potential there is a marked increase in plate current.

If a graph is plotted between plate potential and plate current (for a constant value of filament current) the plate current at first increases slowly for a given increase in plate voltage and when the plate potential is equal to or greater than the ionization

OBSERVATIONS:

S.No.	Plate voltage volts	Plate current μA
1	1	0
2	2	0.5
3	3	1
4	4	2
5	5	3
6	6	3
7	7	4
8	8	5
9	9	6
10	10	8
11	10.5	10
12	11	15
13	11.5	24
14	12	36
15	12.5	out of range
16		
17		
18		
19		
20		

CALCULATIONS:

$$\text{Percentage error} = \left| \frac{\text{Obs. value} - \text{stand. value}}{\text{stand. value}} \right| \times 100$$

$$= \left| \frac{10.5 - 10.29}{10.29} \right| \times 100$$

$$= 0.02 \times 100$$

$$\therefore \text{age error} = 2 \%$$

RESULTS:

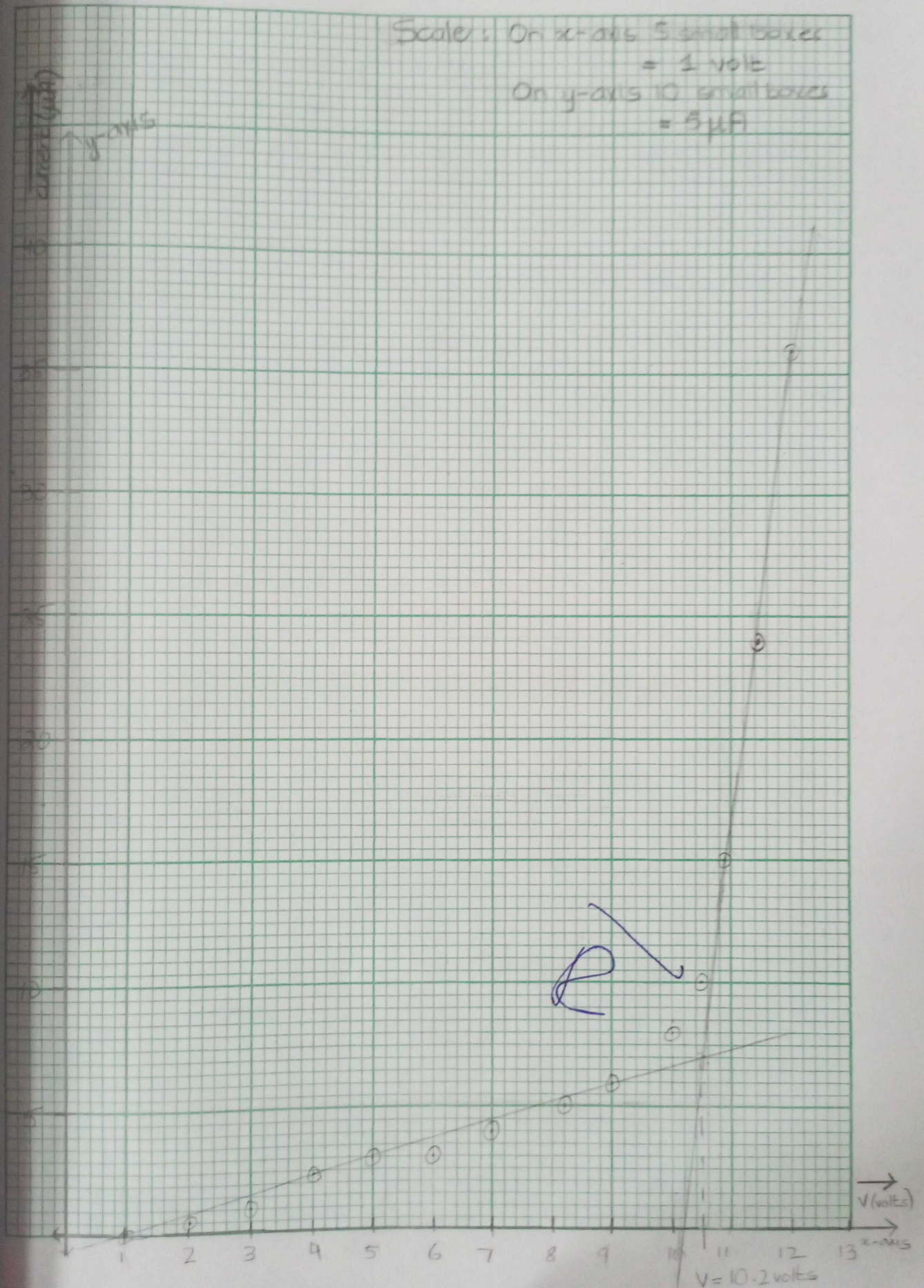
Ionization potential of mercury from the graph = 10.5 Volts

Standard value of ionization potential = 10.29 Volts

Percentage error = 2 %

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Scale: On x-axis 5 small boxes
= 1 volt
On y-axis 10 small boxes
= 5 μA



V = 10.2 volts

Questions

- Q.1 What is the object of your experiment?
- Q.2 What are ions?
- Q.3 What do you mean by ionization?
- Q.4 What is excitation?
- Q.5 What is ionization potential?
- Q.6 Differentiate between excitation and ionization potential?

Ans 1) The object of this experiment is to find I.P of mercury using a gas filled diode.

Ans 2) Ions are atoms or molecules that have gained or loss one or more electron, giving them an electric charge.

- +ve ions (Cations) - Formed when an atom loses an e^-
- -ve ions (Anions) - Formed when an atom gain e^-

Ans 3) Ionization is the process of gaining ~~or~~ losing an electron by an atom or molecule to form ions.

OR

The transfer of electron is ionization.

Ans 5) I.P is ~~the~~ energy required to remove the outermost electron from a neutral atom in its gaseous state

Ans 4) Excitation is the process in which an atom or electron absorbs energy and moves from a lower energy level to higher energy level.

Ans 6) Excitation

- Movement of e^- from L.E.L to H.E.L
- less energy req. compared to ionization
- e^- stays within the atom
- atom remain neutral

I.P

- energy req. to remove e^- from atom
- more energy req. to remove electron
- electron is remove, with +ve charge
- atom loses e^- , become +ve

EXPERIMENT # 03

OBJECT:

To determine the velocity of wave propagation in stretched string by using sonometer.

APPARATUS:

WA-9611 Sonometer, WA-9613 Driver/Detector Coils, Function Generator, Dual trace oscilloscope, Mass and mass hanger.

THEORY:

When two exactly same waves (i-e. same amplitude, frequency and time period) travel in opposite direction with same velocity superimpose on one another the resultant wave obtained is called *standing or stationary wave*.

Hence, we can say that the wave length of standing wave in a vibrating string may be a function of the tension, liner mass density of string, frequency and the velocity of the wave.

If λ is the wavelength of standing wave and v is their velocity, the frequency of the vibration f can be determined by the relation.

$$v = f \lambda \quad (i)$$

The velocity of transverse wave in a string is

$$v = \sqrt{\frac{T}{\mu}} \quad (ii)$$

T is the tension applied to the string and μ is the liner mass density. Substituting the above value of v from (i) to (ii)

$$f \lambda = \sqrt{\frac{T}{\mu}}$$

$$f = \frac{1}{\lambda} \sqrt{\frac{T}{\mu}}$$

whereas the distance l between successive nodes on string give the wavelength of string waves. When string is vibrating in one loop we have

$$l = \lambda/2$$

$$\lambda = 2l$$

and

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

The above equation gives the frequency of transverse wave in a string.

$$\mu = \frac{1}{4L^2} \frac{T}{f^2}$$

PROCEDURE:

1. Set up the Sonometer as shown in Figure 6.
2. Set the bridges 60cm apart. Use any of the included strings and hang a mass of approximately 1 kg from the tensioning lever. Adjust the string knob so that the tensioning lever is horizontal. Position the driver coil approximately 5 cm from one of the bridges and position the detector coil near the center of the wire.
3. Set the signal generator to produce a sine wave.
4. Slowly increase the frequency of the signal driving the driving coil, starting with a frequency of 1 Hz. Determine the lowest frequency for a given tension at which resonance occurs. Record this value in table.
5. Record the string tension (T) in the table. Note the distance between the two wedges.
6. Repeat steps (4) and (5) for different values of tension in the string.
7. Plot a graph between f^2 and tension T.
8. Calculate slope of the graph and find out linear density of string by using

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

9. Take one value of tension of string from graph paper and determine velocity of waves in string by

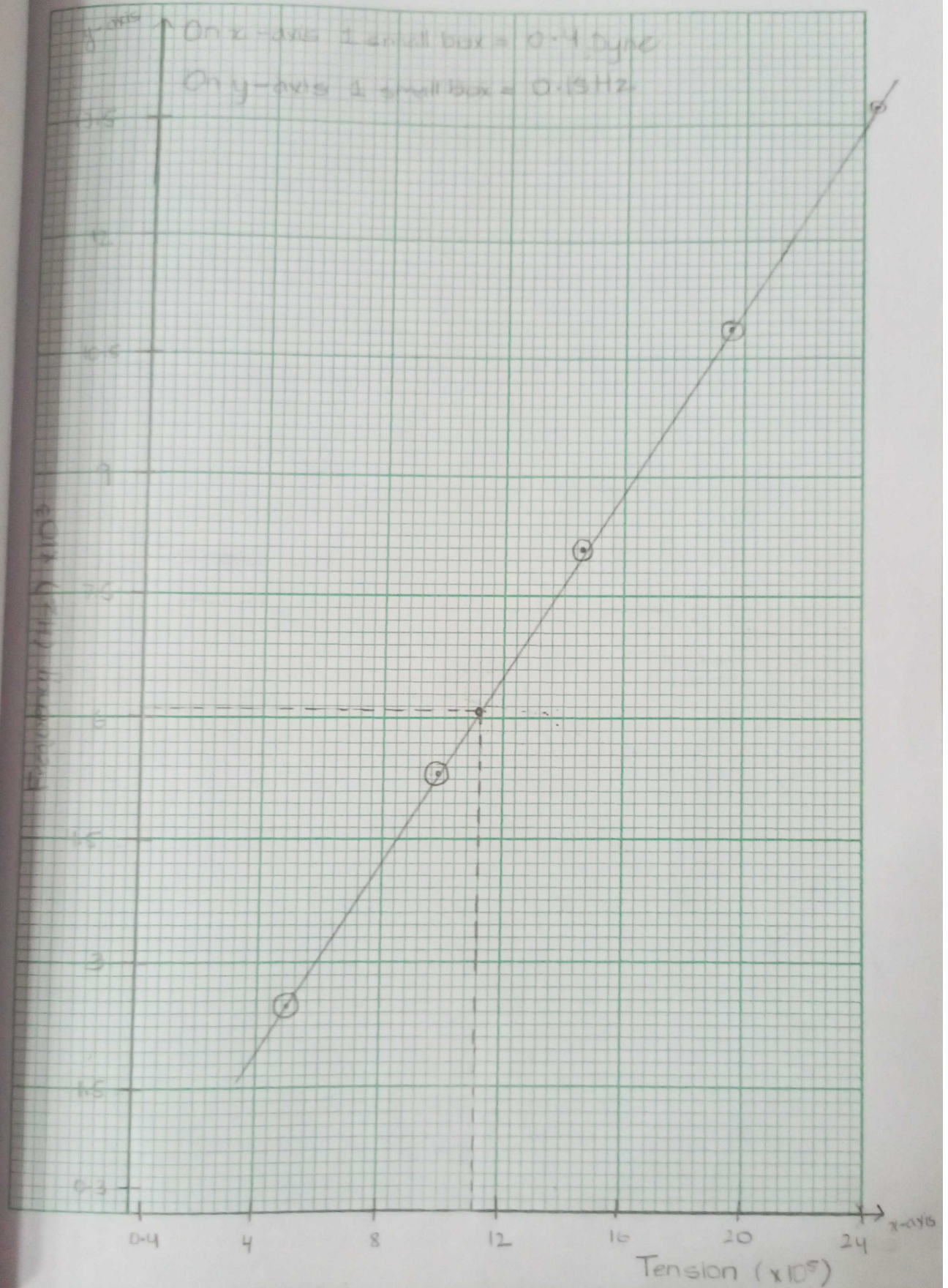
$$v = \sqrt{\frac{T}{\mu}}$$

OBSERVATIONS:

Length of string between two bridges: L = 60 cm

S.No.	Tension Dyne	Fundamental Frequency(f) Hz	f^2 Hz ²
1	$1 \times 980 \times 500 = 4.9 \times 10^5$	50	2.5×10^3
2	$2 \times 980 \times 500 = 9.8 \times 10^5$	74	5.476×10^3
3	$3 \times 980 \times 500 = 14.7 \times 10^5$	90	8.1×10^3
4	$4 \times 980 \times 500 = 19.6 \times 10^5$	104	10.816×10^3
5	$5 \times 980 \times 500 = 24.5 \times 10^5$	116	13.456×10^3

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CALCULATIONS:

$$\mu = \frac{1}{4L^2} \frac{T}{f^2}$$

$$v = \sqrt{\frac{T}{\mu}}$$

FOR μ

$$\therefore \mu = \frac{1}{4L^2} \cdot \frac{T}{f^2}$$

$$\mu = \frac{1}{4(60)^2} \cdot \frac{11.2 \times 10^5}{6.15 \times 10^3}$$

$$\mu = 0.0126 \text{ g/cm}$$

FOR v

$$\therefore v = \sqrt{T/\mu}$$

$$v = \sqrt{\frac{11.2 \times 10^5}{0.0126}}$$

$$v = 9660.91 \text{ cm/sec}$$

R

RESULT:

The velocity of vibrating string at tension 11.2×10^5 dyne is found to be 9660.9 cm/sec

EXPERIMENT # 04

OBJECT:

To determine the horizontal component of Earth Magnetic field strength " H_{ex} " by *Tangent Galvanometer*.

APPARATUS:

Tangent galvanometer, coil, rheostat, reversing key, connecting wires and ammeter.

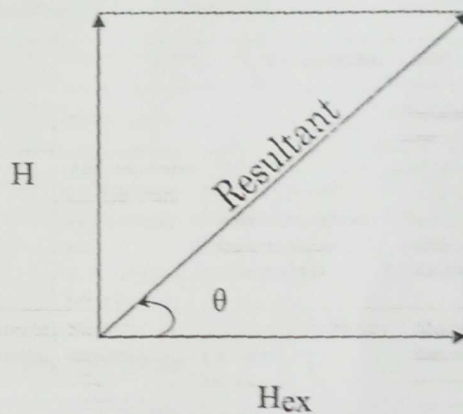
THEORY:

Two magnetic fields can be compared by means of tangent galvanometer, which consist of a magnet pivoted on a vertical axis and carrying a light pointer which can move over a circular scale. Normally one of the field is the earth horizontal component and other field is arranged to be at right angle to this. The pivoted magnet sets itself along the resultant of the two fields at an angle θ to its direction when it is in the earth magnetic field alone. If H_e is magnetizing force of the earth horizontal component and H is the magnetizing force of the other field (Coil).

$$H = H_{ex} \tan \theta \text{-----(1)}$$

The magnetizing force at the center of a circular coil of known radius and known number of turns, when a measured current is passing through the coil, can be calculated. This can be compared by means of tangent galvanometer, with the horizontal component of the earth's magnetizing force, enabling the latter to be determined.

If n is the number of turns in the coil, r meter is its radius, I amp is the current, and H amp per meter the magnetizing force at the center of the coil.



$$H = \frac{nI}{2r} \text{ (amp per meter)}$$

Using equation (1)

$$H = H_{ex} \tan \theta$$

$$H_{ex} = \frac{H}{\tan \theta}$$

$$H_{ex} = \frac{nI}{2r \tan \theta}$$

amp per meter = 0.01256 oersted

PROCEDURE:

1. Connect the circuit as shown in Figure 7.
2. Using leveling screw adjust dial of magnetometer, so that needle could not strike the dial.
3. Adjust the needle of tangent galvanometer so that, the needle is perpendicular to the circular coil.
4. Set rheostat for minimum current (i.e Max resistance), turn on the circuit and give 10° , 15° , 20° , 25° and 30° clockwise deflection to magnetometer needle by increasing the current with rheostat.
5. Reverse the keys of reversing switch and repeat the step 4 in anticlockwise direction.
6. Note down the corresponding current for each deflection.
7. Plot a graph between I and $\tan\theta$, which will be a straight line of slope $I/\tan\theta$.
8. Knowing also N and r , H_{ex} can be calculated.

OBSERVATIONS:

No. of turns in the coil = $n = 50$ turns
Radius of the coil = $r = 0.0715$ m

S.No.	Angle θ Deg	Current in mA			$\tan\theta$
		Clock Wise Reading I_1 mA	Anti-Clock Wise Reading I_2 mA	Mean $I = (I_1 + I_2)/2$ mA	
01	15	21	20	20.5	0.27
02	20	28	26	27	0.36
03	25	36	36	36	0.47
04	30	45	42	43.5	0.58
05	35	56	53	54.5	0.70

CALCULATION:

$$H_{ex} = \frac{nI}{2r \tan \theta} \text{ amp per meter}$$

1 amp per meter = 0.01256 oersted

$$H_{ex} = \frac{(50)(40 \times 10^{-3})}{2(0.0715)(0.52)}$$

$$H_{ex} = 26.89 \text{ amp per meter}$$

$$H_{ex} = 26.89 \times 0.0125$$

$$H_{ex} = 0.33 \text{ oersted}$$

RESULT:

The value of H_c is found to be = 0.33 (Oersted)

PRECAUTIONS:

1. Connections should be tight.
2. Angle should be adjusted with care.
3. Dial of galvanometer should be leveled properly.
4. Keep Metallic Material away from magnetic dial.

CIRCUIT DIAGRAM:

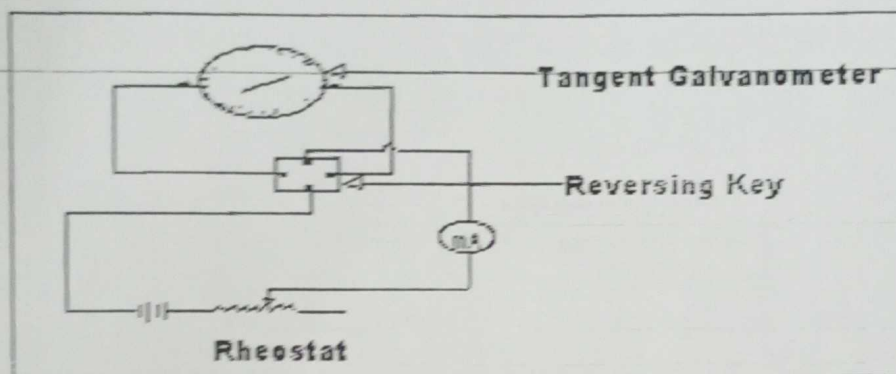


Figure 7: "H" Tangent Galvanometer

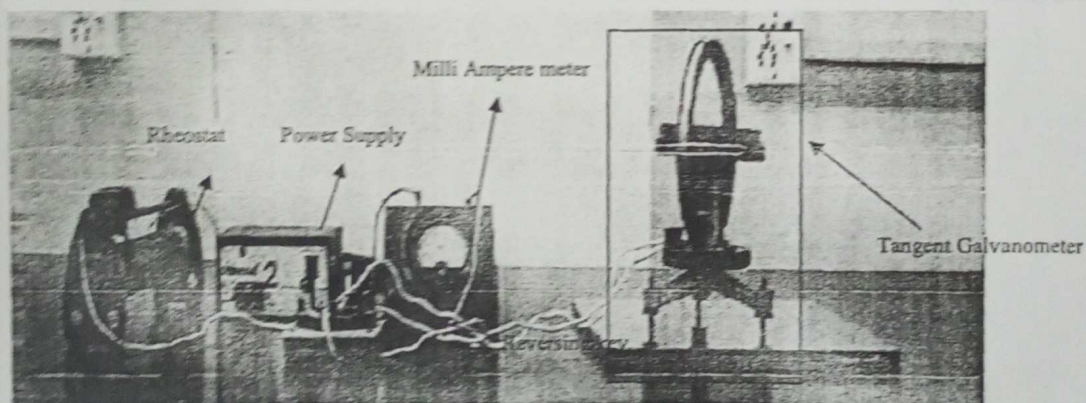
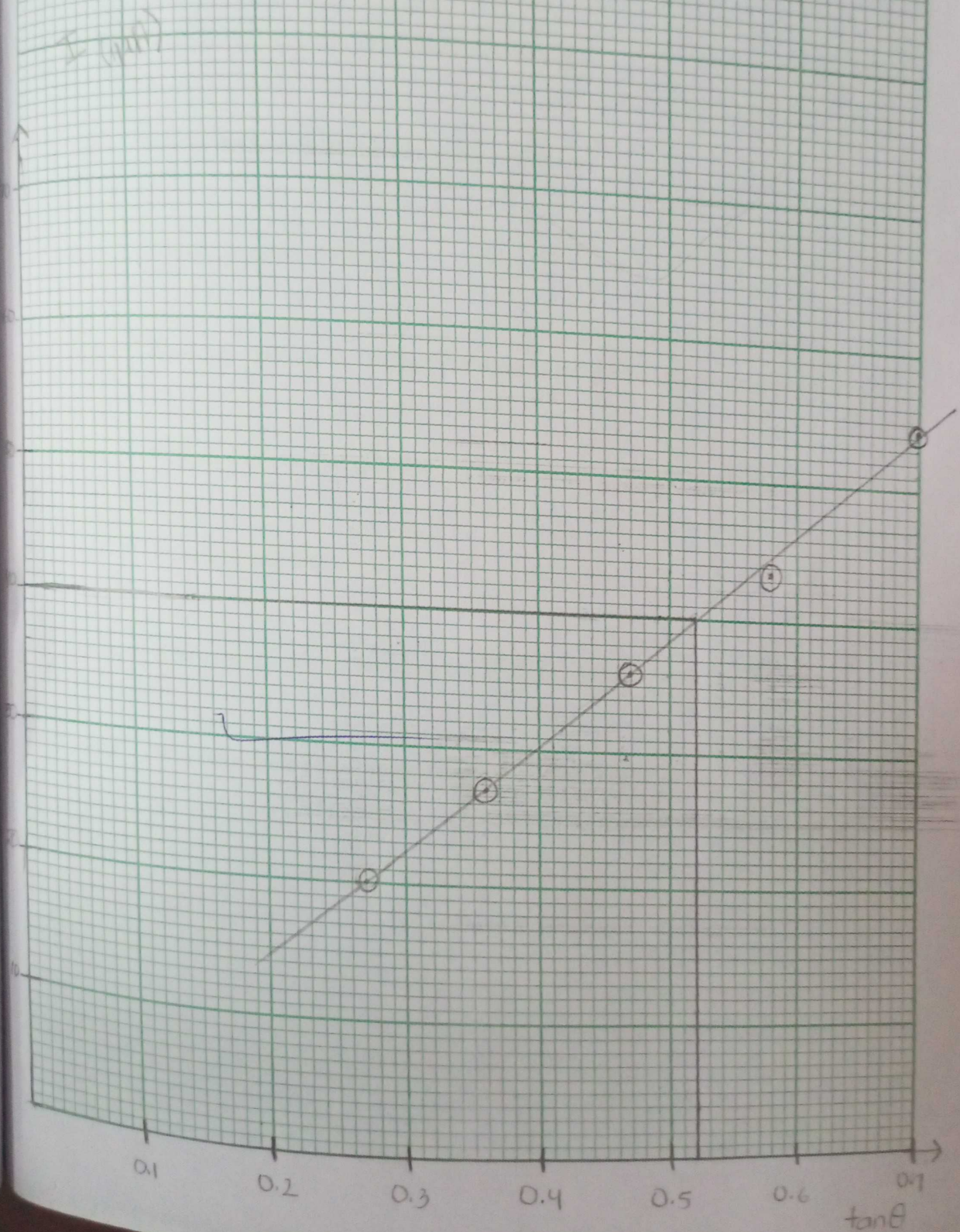


Figure 8: Photograph of the apparatus

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On x-axis 10 small boxes = 0.1°C

On y-axis 10 small boxes = $10\ \mu\text{A}$



EXPERIMENT # 05

OBJECT:

To study the characteristics of an acceptor circuit and determine unknown inductance.

APPARATUS:

Resistance, capacitors, inductor, a frequency generator/oscillator, an Oscilloscope, connecting wires, A.C. supply etc.

THEORY:

Acceptor Circuit:

When the resistor R, inductor L and capacitor C are connected in series with a source of emf E, the circuit is known as the RLC-series or series resonant or series tuned circuit as shown in figure.

The total opposition offered by the RLC circuit in flow of current is called impedance Z. it depends on frequency of AC signal applied and given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

Where $X_L = \omega L$ and $X_C = \frac{1}{\omega C}$ are inductive and capacitive reactance of inductor and capacitor respectively

In A.C. circuits the voltage and the current are usually not in phase. Across the inductor, the current lags behind the voltage by 90° , whereas across the capacitor, the current leads the voltage by 90° . But across the resistor the voltage and current both are in phase. At particular frequency the capacitive reactance X_C and the inductive reactance X_L become equal and cancel effect of each other and circuit behave as purely resistive circuit. Under this condition, the voltage and current are in phase, even though the circuit consists of L, C and R. This phenomenon is called resonance and the frequency is known as resonant frequency.

$$X_L = X_C$$

$$\omega L = \frac{1}{\omega C}$$

$$\omega = 2\pi f = \frac{1}{\sqrt{LC}}$$

$$f = \frac{1}{2\pi\sqrt{LC}}$$

This is an acceptor circuit, that means it allows maximum current to flow through it at this particular (resonant) frequency and at all other frequencies it allows less current.

EXPERIMENT 1

Capacitance of capacitor $C = 0.001 \mu F$

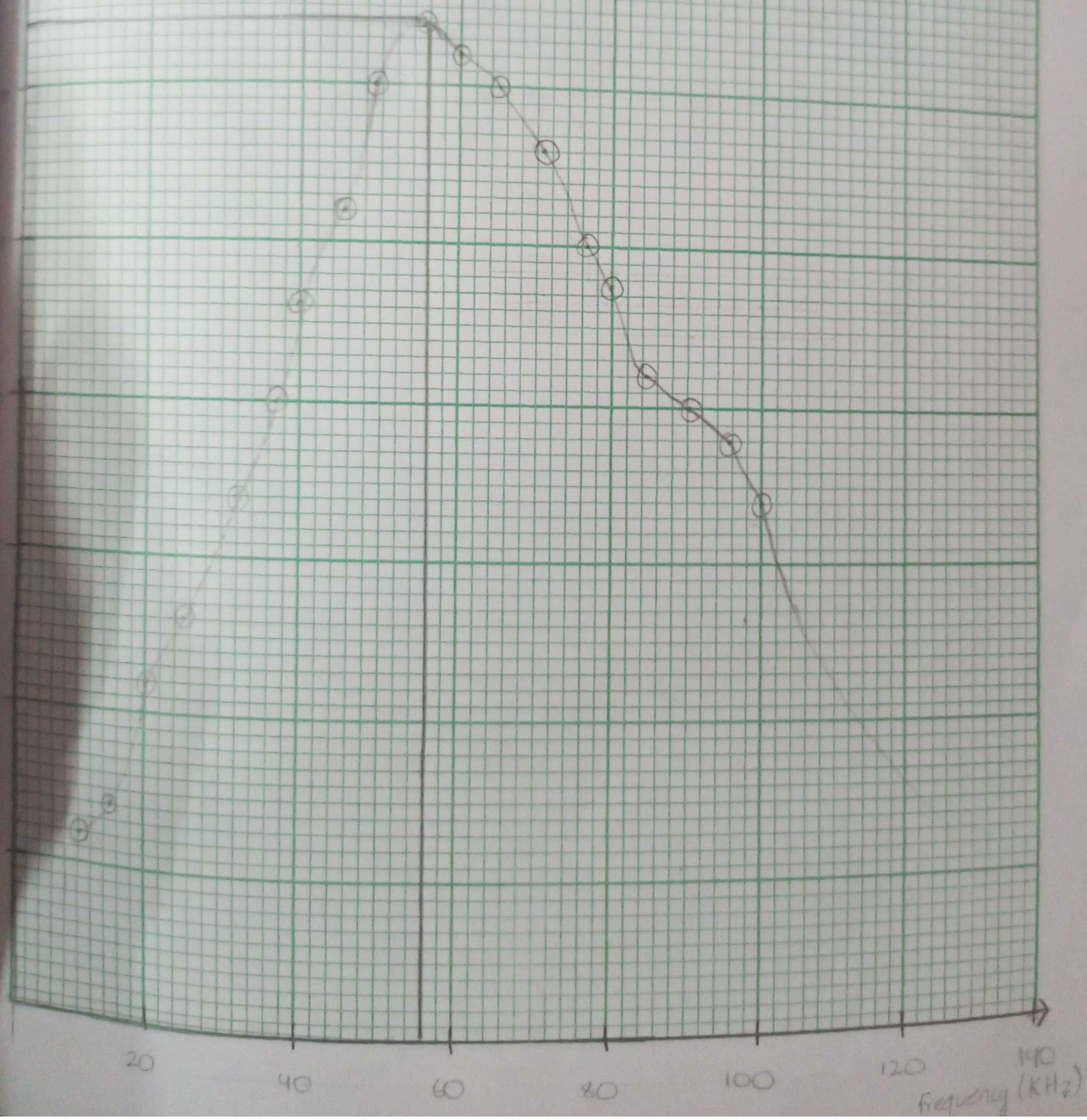
S. No.	Frequency kHz	Amplitude of Signal mV
1	10	5
2	15	8
3	20	11
4	25	13
5	30	14
6	35	20
7	40	23
8	45	25
9	50	26
10	55	27
11	60	28
12	65	29
13	70	30
14	75	31
15	80	32
16	85	33
17	90	34
18	95	35
19	100	36
20		

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On x-axis 10 small boxes = 20 kHz

On y-axis 10 small boxes = 5 amplitude

(w)



CALCULATIONS:

From Graph:

1- Value of resonant frequency $f = \dots 55 \dots$ kHz.

$$L = \frac{1}{4\pi^2 f^2 c}$$

$$L = \frac{1}{4\pi^2 (55 \times 10^3)^2 (0.001 \times 10^{-6})}$$

$$L = 8.37 \times 10^{-3} \text{ Henry}$$

RESULT:

1. Characteristics of a acceptor circuit studied. It is seen that at resonant frequency the output signal is maximum / minimum.
2. The value of inductance is found $L = 8.37 \times 10^{-3}$ Henry.

PRECAUTIONS AND SOURCES OF ERROR:

1. Before switching on, get connections checked by teacher.
2. The amplitude of the signal at resonant frequency may not be exactly zero, which is due to the presence of the resistance.
3. Choose proper combinations of L, C & R

EXPERIMENT # 06

OBJECT:

To determine the unknown high resistance by Neon Flash lamp Apparatus.

APPARATUS:

Neon lamp, known and unknown high resistance, a capacitor of suitable value, stops watch and power peck etc.

THEORY:

Flashing and quenching of a neon bulb. A neon bulb is placed in parallel with a capacitor and connected to D.C. supply which can be continuously increased from 0 to 150 volt through a high resistance of about $1\text{M}\Omega$. The voltage is slowly increased to a value say V_1 when the lamp flashes and begins to glow. As soon as the neon lamp flashes, it becomes conducting and the capacitor begins to discharge through it. It continues to do so until the extinction (or quenching) potential V_2 is reached when the neon lamp ceases to glow and stops conducting. The capacitor then again begins to charge till the flashing potential V_1 is reached when again the lamp flashes and begins to glow. The process is repeated. During the time the capacitor is charging the neon lamp does not glow. In other words, the total time t between two consecutive flashes is equal to the time taken by the voltage first to fall from the flashing potential V_1 to quenching potential V_2 (discharge) and then to rise from V_2 to V_1 (charging). This flashing and quenching time can be determined by noting the time taken by the lamp to produce say 20 consecutive flashes and quenches. If t_1 is the time taken by the capacitor voltage to fall from V_1 to V_2 and t_2 is the time taken by the voltage to rise from V_2 to V_1 , then

V_2 for charging of capacitor, $V_2 = V_1 e^{-t_1/RC}$

$$t_1 = -CR \log_e \frac{V_2}{V_1}$$

V_2 for discharging of capacitor, $V_2 = V_1 (1 - e^{-t_2/RC})$

$$t_2 = -CR \log_e \left(1 - \frac{V_2}{V_1}\right)$$

$$t = t_1 + t_2 = R \left[-C \log_e \frac{V_2}{V_1} - C \log_e \left(1 - \frac{V_2}{V_1}\right) \right]$$

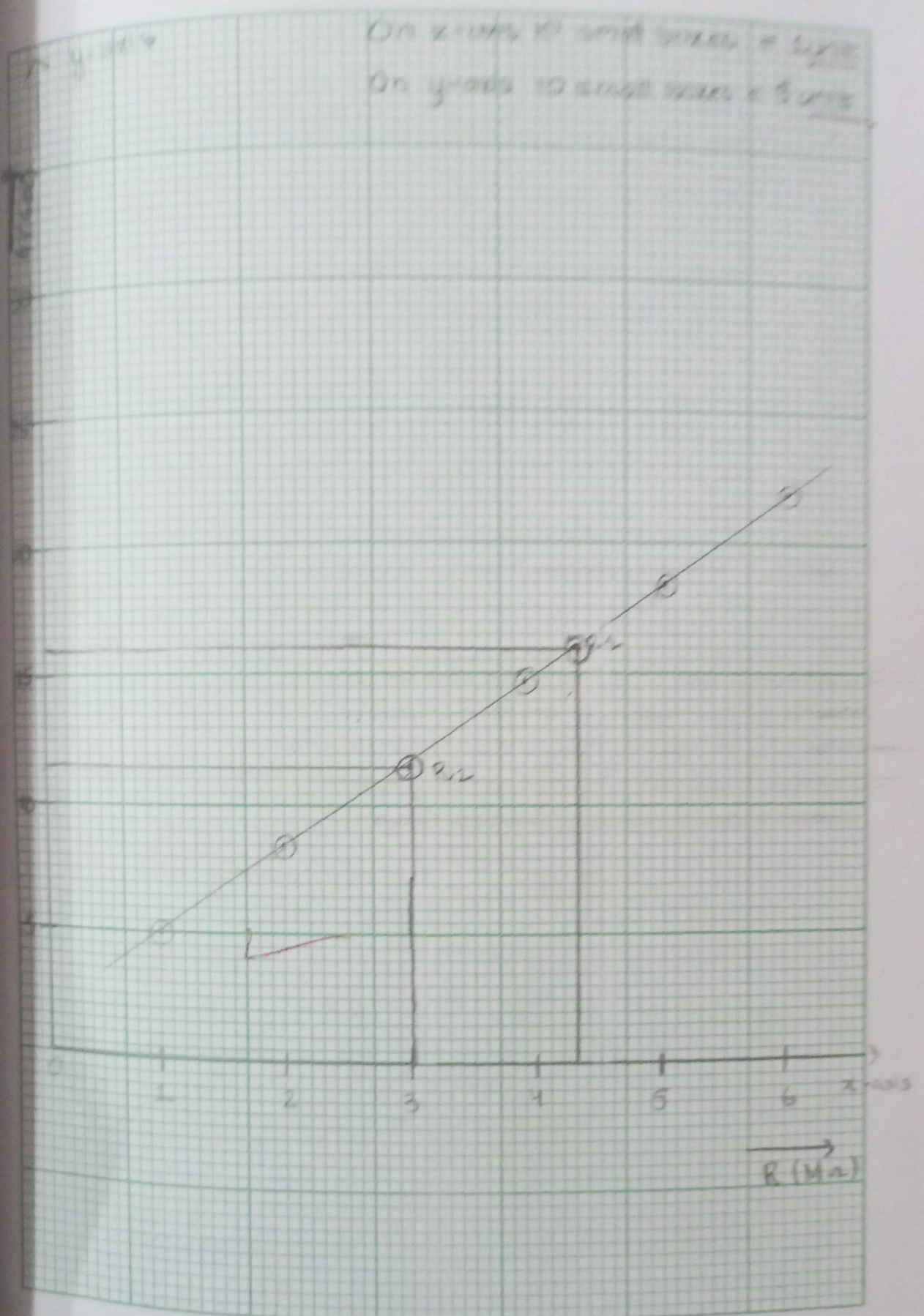
As C , V_2 and V_1 have constant fixed values, the function within brackets is a constant, and denoted by k

$$\therefore t = kR$$

$$t \propto R$$

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On x-axis is unit vector $\hat{x}$ & $\hat{y}$
 On y-axis is unit vector $\hat{y}$ & $\hat{z}$



$$R_1 = 4.4 - 1 = 3.4 \text{ Ma}$$

$$R_2 = 3 - 1 = 2 \text{ Ma}$$

OBSERVATIONS:

The capacitance of the capacitor = 1 μF

S.No.	Resistance R M Ω	Time for Ten flashes Seconds			Mean Time t Seconds	Time for one flash T=t/10(sec)
		t ₁	t ₂	t ₃		
1	1 M Ω	4.75	5.78		5.265	0.5265
2	2 M Ω	8.1	8.31		8.205	0.8202
3	3 M Ω	11.40	11.11		11.255	1.1255
4	4 M Ω	14.54	14.80		14.67	1.467
5	5 M Ω	18.41	18.65		18.03	1.803
6	6 M Ω	22.52	22.80		22.80	2.280
7	R ₁ +1	16.20	15.80		16.00	1.600
8	R ₂ +1	12.22	10.48		11.45	1.145
9						
10						

RESULT:

From Graph:

The value of unknown resistance from the graph R₁ = 3.4 M Ω .
The value of unknown resistance from the graph R₂ = 2 M Ω .

PRECAUTIONS AND SOURCES OF ERROR:

1. Don't touch naked wires.
2. The applied voltage must be uniform and must be kept constant throughout the experiment.
3. Note the time for flashes carefully.

CIRCUIT DIAGRAM:

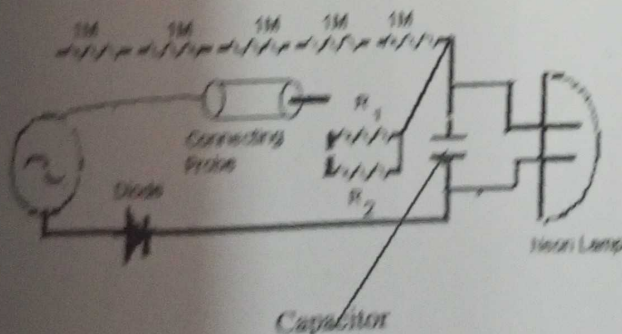


Figure 12: Circuit Diagram

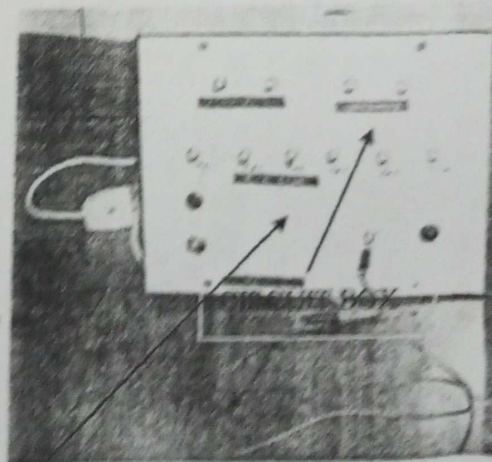


Figure 11: Photograph of the apparatus